

THÉO MATRICON

AI Data & Software Engineer

Large-Scale Data Pipelines · AI Infrastructure · Performance Engineering

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PROFILE

Computer Science PhD and software engineer building **large-scale data pipelines, AI research infrastructure, and high-performance software**. Experienced in turning experimental ideas into reliable Python/Linux systems, distributed workflows, and reproducible evaluations. Interested in data ingestion, audio and language AI, and infrastructure that shifts the cost/throughput/quality frontier for model training.

IMPACT AT A GLANCE

200M+ repositories processed	200k+ CPU hours orchestrated	50% lower benchmark cost	4×+ CUDA speedup
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PORTFOLIO

Agent Analytics Built a large-scale ingestion and analysis pipeline for commits, files, and pull requests across 200M+ GitHub repositories. Python · SQL · Linux · distributed processing · APIs	DATA SYSTEMS	Efficient Evaluation Designed comparison workflows that cut large-scale benchmarking costs by 50% while preserving reliable rankings. Python · automation · experimental design · reproducibility	INFRASTRUCTURE
GPU-Accelerated Synthesis Delivered a 4×+ speedup for a state-of-the-art MBA synthesizer by optimizing its CUDA implementation and execution strategy. CUDA · C++ · profiling · program synthesis	PERFORMANCE	DeepSynth2 & EcoSearch Designed a 20k+ LOC framework and faster search algorithms for repeatable AI training/evaluation experiments. Python · PyTorch · Cython · search · ML workflows	AI SOFTWARE

EXPERTISE

Languages	Python · Bash · C/C++ · CUDA · SQL · JavaScript
Technologies	Linux · Docker · PyTorch · Git · GitHub Actions · REST/GraphQL · pytest/ruff/pre-commit
Systems	data ingestion and processing · AI infrastructure · HPC/cluster execution · testing · performance optimization
Methods	optimization · search algorithms · machine learning · structured evaluation · reproducible experiments
Collaboration	async writing · technical supervision · research roadmapping · software engineering teaching

EXPERIENCE

Scientist INRIA · Rennes, France	2025–PRESENT AI for Software Engineering
- Built AI/data systems spanning repository-scale ingestion, CUDA acceleration, and cluster-scale evaluation; published in top venues (AAAI, TOSEM, MSR). - Orchestrated experiments across 50+ nodes and 200k+ CPU hours; supervised 3 interns and advised 5 junior researchers through autonomous, written work.	

PhD Scientist University of Bordeaux / LaBRI · Bordeaux, France	2021–2024 Program Synthesis & Code Generation
- Designed ML-powered synthesis systems and a 20k+ LOC Python/PyTorch framework for repeatable AI experiments. - Developed production-quality Linux software with Python, C/C++, Cython, SQL, testing, profiling, and automation; supervised 7+ interns and 2 engineers.	

EDUCATION

PhD in Computer Science · University of Bordeaux Program Synthesis and Code Generation	2021–2024
Engineering Degree / MSc in Computer Science · ENSEIRB-MATMECA Artificial Intelligence	2018–2021